

LUNABOT: AUTONOMOUS NAVIGATION ROBOT FOR LUNAR HABITAT

Abstract

LunaBot is an autonomous robotic system designed for navigation in lunar habitats using sensors, obstacle detection and AI-based decision making.

Introduction

The project focuses on autonomous navigation, monitoring and safe movement in lunar environments.

Objectives

- Autonomous Navigation
- Obstacle Detection
- Dashboard Monitoring
- Database Management

Modules

User Authentication, Navigation Module, Obstacle Detection, Dashboard Module, Database Module.

Project Development Schedule

1. Project Title Finalization
2. Requirement Gathering
3. Objective Definition
4. User & Module Identification
5. Use Case Diagram Preparation
6. Database Requirement Analysis
7. ER Diagram Design
8. Database Schema Creation
9. UI Wireframe Design
10. Login & Dashboard UI Design
11. Navigation & Form Design
12. Design Review
13. Frontend Environment Setup
14. Login Page Development
15. Registration Page Development
16. Dashboard Development
17. CRUD Form Development
18. Table & Search Features
19. Frontend Testing
20. Spring Boot Project Setup
21. Database Connectivity
22. Entity & Repository Creation
23. REST API Development
24. Authentication Module
25. Backend Business Logic
26. API Testing using Postman
27. Frontend & Backend Integration
28. Bug Fixing & Validation
29. Documentation & PPT Preparation
30. Final Demo & Submission